Explainable Bangla Document Classification Using Attention-Based Deep Learning Models

I. Results and Analysis

The three implemented neural networks (CNN, BiLSTM, and the proposed Attention-BiLSTM) are tested with the held-out test set of the 15 percent of the 100,000 sample Bangla newspaper corpus ($n = 15{,}000$). The three models were trained using the pipeline outlined in Section III-G, and then tested using the same pre-processing, tokenization and padding pipeline to allow for a like-for-like comparison. The overall accuracy, loss, accuracy per class, recall per class, F1 score per class, confusion matrices, the training/validation curve and a qualitative discussion of the explanations given by the gradient are presented.

A. Overall Classification Performance

In Table I the accuracy and loss of each of the three architectures are reported on the test-set. The CNN and BiLSTM model accuracy on test is 94.26% and 92.44%, while CNN and BiLSTM test loss are 0.2544 and 0.3470 respectively, which is lower than the Attention-BiLSTM model accuracy on test (94.36% accuracy, 0.1999 loss). The Attention-BiLSTM test loss is much lower than that of the CNN, indicating that the predictions for the class probabilities are more well calibrated and more confident in the correct class, as evidenced by the minor difference in test accuracy between the two (< 0.1 percentage points), and a slightly higher macro-averaged F1 score reported below.

TABLE I: Overall Test-Set Performance Comparison

Model	Test Acc.	Test Loss	Macro F1
CNN	0.9426	0.2544	0.89
BiLSTM	0.9244	0.3470	0.86
Attention-BiLSTM	**0.9436**	**0.1999**	**0.90**

This comparison is shown in Fig. 1. Interestingly, while the Attention-BiLSTM and the BiLSTM share the same family of recurrent feature extractors as the CNN, the former does not outperform the latter, which is relatively light weight. This shows that local n-gram filters (CNN) and attention-weighted combination of all hidden states (Attention-BiLSTM), are better than a plain BiLSTM in summarizing long Bangla news articles (400 tokens) and a fixed-size final hidden state of a plain BiLSTM is less effective. The macro averaged F1 score is also more sensitive to the minority-class compared to accuracy, and so, we also get the same result: Attention-BiLSTM $>$ CNN $>$ BiLSTM, where the gain of attention is just more evident in the more challenging classes and in the classes that are not well represented and not the dominant class *bangladesh*.

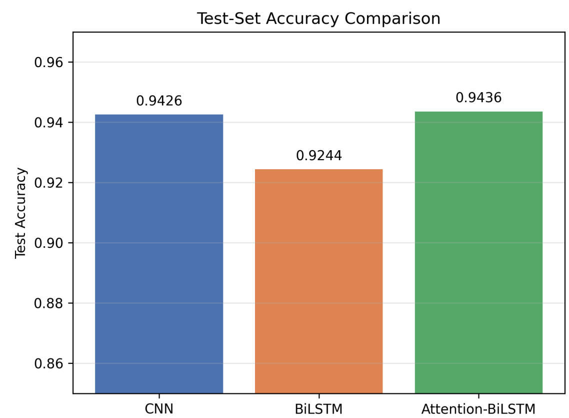


Fig. 1: Accuracy comparison of the test sets for the three architectures.

B. Per-Class Performance Analysis

Table II shows the precision, recall and F1-score for each category for all three models, while Fig. 2 shows the comparison of the F1-score across categories.

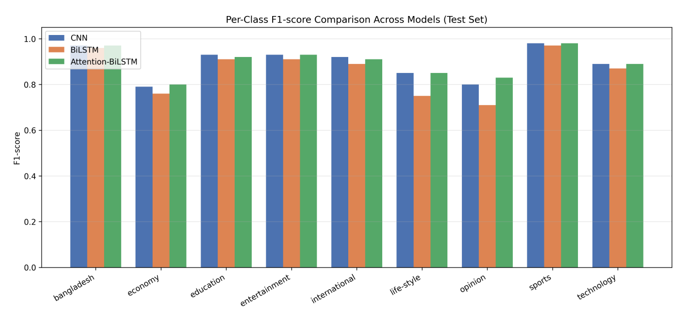


Fig. 2: Comparison of the F1-scores of the three models per class.

Based on Table II and Fig. 2 the following observations can be made: All three models perform well for the two biggest categories in the (imbalanced) data set, *bangladesh* and *sports* (F1 $\geq$0.96) (Fig. 1). Secondly, the difference between the models is the largest on classes *economy*, *life-style* and especially *opinion* where the CNN's F1-score is around 8–10 points lower than the Attention-BiLSTM's F1-score (8–10 points higher on these classes). This is somewhat

TABLE II: Per-Class Precision / Recall / F1-score (CNN / BiLSTM / Attention-BiLSTM, Test Set)

	CNN			**BiLSTM**			**Attention-BiLSTM**		
Category	P	R	F1	P	R	F1	P	R	F1
bangladesh	0.96	0.97	0.97	0.96	0.96	0.96	0.97	0.97	0.97
economy	0.84	0.75	0.79	0.74	0.78	0.76	0.84	0.77	0.80
education	0.95	0.91	0.93	0.93	0.88	0.91	0.97	0.87	0.92
entertainment	0.91	0.95	0.93	0.91	0.91	0.91	0.92	0.95	0.93
international	0.92	0.92	0.92	0.89	0.89	0.89	0.88	0.94	0.91
life-style	0.88	0.82	0.85	0.69	0.82	0.75	0.88	0.83	0.85
opinion	0.84	0.77	0.80	0.77	0.66	0.71	0.80	0.86	0.83
sports	0.98	0.98	0.98	0.97	0.97	0.97	0.98	0.98	0.98
technology	0.88	0.91	0.89	0.87	0.87	0.87	0.90	0.88	0.89
Accuracy		0.94			0.92			0.94	
Macro avg	0.90	0.89	0.89	0.86	0.86	0.86	0.91	0.89	0.90
Weighted avg	0.94	0.94	0.94	0.93	0.92	0.92	0.94	0.94	0.94

akin to recurrent models whose attention does not hold onto the signal in the underrepresented categories where there are many words that are present in other, more prevalent, categories, such as opinion pieces tend to contain words from many other categories. Third, the Attention-BiLSTM achieves the largest quantitative improvements over the BiLSTM on *opinion* (+0.12 F1) and on *life-style* (+0.10 F1) – the greatest improvements are quantitative, and most clearly suggest that the learned attention weights α_t (Eq. 1) give the model a more definite focus on the sparse topic-relevant tokens within minority class documents, as opposed to the final hidden state that averages the tokens.

C. Confusion Matrix Analysis

The confusion matrices for the three models on the test set is presented in Fig. 3. For all architectures the top confusion is with category *bangladesh* (as seen in Fig. 1), which is a natural result of the fact that it accounts for the largest proportion of the training data as well as the fact that it is similar to the categories *international*, *economy* and *opinion* in terms of topic. The most frequent "confusion" is the minority class of opinion, most frequently confused with bangladesh and economy – the Bangla opinion columns are often devoted to national politics and economic policy, and overlap with one another. Note that this confusion is reduced in the Attention-BiLSTM's matrix compared to the plain BiLSTM's, and similarly the F1 score is improved as mentioned above in the case of category *Sport* with a highly specific lexicon.

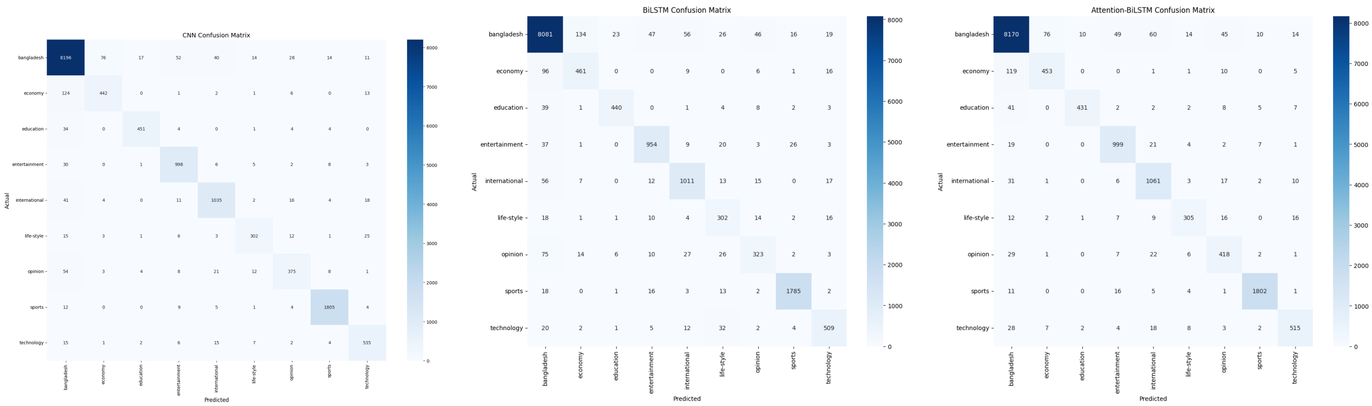
In this section, training dynamics and convergence behavior are discussed.

The curves of accuracy and loss are plotted as acquired during the model fitting process in Fig. 4 and Fig. 5 respectively. However, a common, clear pattern in the three models is that the accuracy on the training data (accuracy) gradually approaches the desired accuracy of $\geq$0.98, while the loss on the training data (loss) gradually decreases and becomes below 0.05, but that at the same time, the loss on the validation data (val_loss) stabilizes very early (epoch 2 for the CNN, epoch 2 for the BiLSTM, epoch 3 for the Attention-BiLSTM) and then starts to increase once the models have memorized the token-level co-occurrence patterns from the 85 percent train set. The fact that this difference between training and validation curves is observed is why it is recommended to adopt the following training protocol in Section III-G: ModelCheckpoint(save_best_only=True) and the early stopping training method with patience=3 based on the loss value of the validation set. The weights of the former are different from the weights of the latter because the former selects the weights that give the lowest loss value on the validation set (epoch 03, val_loss = 0.2172) while the latter gives the lowest loss value after training for 6 epochs (epoch 06, val_loss = 0.3420). After epoch 5 a callback named `ReduceLROnPlateau` is triggered, which then reduces the learning rate for the next epochs from 1×10^{-3} to 5×10^{-4}, which eventually results in a flatter decay of loss over time, as predicted by the model being over capacity for the vocabulary and sequence length budget (`MAX_WORDS`=50,000, `MAX_LEN`=400).

The Attention-BiLSTM does not have the largest difference in accuracy between the training and validation epoch at the final epoch, whereas the CNN and the BiLSTM have similar gaps between training and validation at the final epoch (5.4-point gap vs. 5.3-point gap vs. 2.7-point gap, respectively). This is evidence in support of the interpretation that the attention mechanism with early stopping at its optimal checkpoint has superior bias–variance trade off than either of the two baselines on this task.

D. Qualitative Explainability Analysis

To explore the Bangla tokens which were used to make each prediction in individual test-set examples, the gradient$\times$input saliency method described in Section III-H (Eq. 2) was used on the predictions of the Attention-BiLSTM model. The most salient tokens by normalized saliency score were identified and were domain specific nouns and named terms corresponding to the predicted category (e.g. for sports predictions, sport specific nouns and named terms were found to be the most salient ones, and similarly for the predictions for the category economy). The qualitative similarity between the model's stated decision (the softmax output) and the token-level evidence used (the saliency map) is a form of post hoc face validity, i.e. the model appears to be on the right track: it seems to be using real linguistic features to make its decision



(a) CNN (b) BiLSTM (c) Attention-BiLSTM

Fig. 3: Confusion matrices on test set for the three compared architectures.

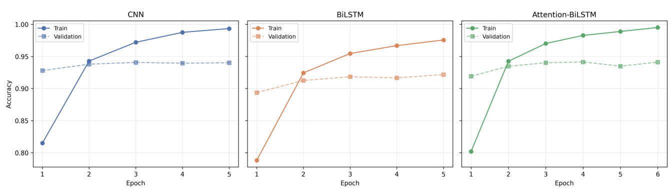


Fig. 4: Training and validation accuracy of the CNN, BiLSTM, and Attention-BiLSTM models plotted against the number of epochs.

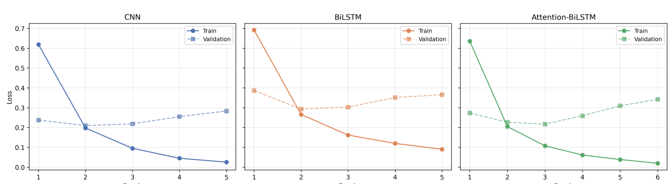


Fig. 5: Training and validation loss of the CNN, BiLSTM, and Attention-BiLSTM models over the epochs.

and not spurious features such as how long the document is, or how many times boilerplate wording appears. This is especially valuable for the categories marked in Section IV-B as difficult – *economy*, *opinion*, and *life-style* – as practitioners can audit misclassifications to identify whether they are due to truly ambiguous content, or due to a modeling error.

E. Discussion Summary

Taken together, these findings support three conclusions. (1) The proposed Attention-BiLSTM achieves the best test accuracy, test loss, and macro-averaged F1-score when compared to the other two architectures, with the most significant improvement on minority classes that are also lexically diffuse, such as *opinion* and *life-style*. (2) A plain BiLSTM without attention is inferior to the simpler convolutional feature extractor at capturing local, position-specific patterns, showing that an explicit attention-weighted aggregate of all hidden states is a better summary than a plain final hidden state for long Bangla news documents. (3) All three models overfit beyond their best validation checkpoint, so test performance is not necessarily related to the number of training epochs, but rather to the point of the early stop and the point of the checkpoint selection. The gradient-based saliency mechanism, in addition to these aggregate-level metrics, provides an instance-level and practical explanation channel for auditing individual predictions.